

Indian Statistical Institute

BSDS IInd Year

Academic Year 2025 - 2026: Semester I

Course: Probability II

Instructor: Antar Bandyopadhyay

Assignment # 12

Date Given: November 12, 2025

Date Due: November 21, 2025
Total Points: 10

1. Suppose X and Y are *i.i.d. standard normal random variables*. Then show that $T = X/Y$ is a random variable which is *Cauchy* distributed. Hence or otherwise show that *Student's* t_1 is also Cauchy distribution.
2. Suppose $T_n \sim t_n$, where t_n is the *Student's t-distribution* with n degrees of freedom, then show that $T_n \xrightarrow{d} \text{Normal}(0, 1)$ as $n \rightarrow \infty$.
3. Find an appropriate centering μ_k , and a scaling σ_k , such that,

$$\frac{X_k - \mu_k}{\sigma_k} \xrightarrow{d} \text{Normal}(0, 1), \text{ as } k \rightarrow \infty,$$

where $X_k \sim \text{Negative-Binomial}(k, p)$ and $0 < p < 1$. Recall X_k is the *number of failures before k successes* from *i.i.d. Bernoulli (p) trials*.

4. In the context of **Problem # 3**, find a transformation g , such that the *asymptotic* variance of $g(X_k)$ is free of p .